

NATURAL SCENE IMAGE CLASSIFICATION

Deep Learning - Transfer Learning - Explainable AI - Deployment

Final Computer Vision Project Report

A concise end-to-end report covering problem definition, data preparation, model development, evaluation, explainability, and deployment.

6

SCENE CLASSES

92.17%

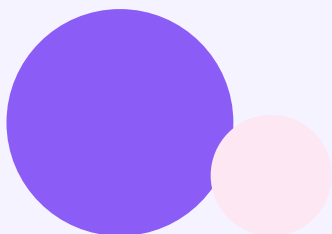
TEST ACCURACY

92.35%

MACRO F1

Final Model	Fine-Tuned MobileNetV2
Input	224 × 224 × 3 RGB images
Deployment	Reusable inference + Streamlit interface

Buildings • Forest • Glacier • Mountain • Sea • Street



1. Problem, Dataset & Data Pipeline

Problem. The project solves a six-class natural-scene image classification task. Given one RGB photograph, the system predicts exactly one label: buildings, forest, glacier, mountain, sea, or street. Because each image belongs to one of more than two mutually exclusive categories, the task is a **multi-class classification** problem.

Dataset. The project uses the Intel Natural Scene Classification dataset. The development workflow inspects the actual files and class distribution rather than assuming fixed counts. The final transfer-learning pipeline uses 11,929 training images, 2,105 validation images, and 3,000 held-out test images.

Image EDA - Key Findings

- The dataset contains six semantically distinct classes, but several categories share visual structure.
- **Mountain and glacier** are naturally difficult to separate because both can contain rock, snow, ice, sky, and high-elevation terrain.
- **Buildings and street** also overlap because urban street scenes frequently contain prominent building structures.
- Image-size, pixel-value, class-distribution, and representative-sample checks were performed before modeling.

Preprocessing & Augmentation

The baseline CNN uses 150×150 RGB inputs, while MobileNetV2 uses 224×224 RGB inputs. The final exported transfer model contains its required MobileNetV2 rescaling internally. Training augmentation is intentionally lightweight: horizontal flipping, small rotations, and small zoom transformations. Augmentation is applied only to training data; validation and test data remain unaugmented to preserve fair evaluation.

Split	Purpose	Model-selection rule
Train	Learn model parameters	Augmentation allowed
Validation	Tune and select model	No augmentation
Test	Final unbiased evaluation	Untouched until final selection

Leakage control: model selection is based on validation performance; the test set is reserved for the final evaluation only.

2. Model Development & Training Strategy

Baseline CNN

A custom CNN was developed first to establish a transparent benchmark. Its architecture progressively learns spatial features using convolutional blocks, pooling, Batch Normalization, dropout, global average pooling, and a six-unit Softmax output layer. The baseline contains **423,302 trainable parameters** and uses $150 \times 150 \times 3$ inputs.

Transfer Learning with MobileNetV2

The stronger model uses **MobileNetV2 pretrained on ImageNet**. Training was performed in two controlled stages. First, the convolutional backbone was frozen and only the custom classification head was trained. Second, a limited set of upper backbone layers was unfrozen for fine-tuning with a much smaller learning rate, while Batch Normalization layers remained frozen.

Stage	Strategy	Outcome
Feature extraction	Frozen MobileNetV2 + new classifier head	Best val. acc. 91.83%; val. loss 0.2347
Controlled fine-tuning	Upper layers trainable; LR = 1e-5	Selected val. acc. 92.78%; val. loss 0.2088

Training Controls

- **EarlyStopping** stops unnecessary epochs and restores the best weights.
- **ReduceLROnPlateau** reduces the learning rate when validation improvement stalls.
- **Dropout and augmentation** improve generalization and reduce reliance on narrow visual patterns.
- **Controlled fine-tuning** adapts pretrained high-level features without aggressively overwriting useful ImageNet representations.

Overfitting assessment. The final transfer model shows no clear evidence of substantial overfitting: the selected validation accuracy is 92.78% and the held-out test accuracy is 92.17%, a gap of only about 0.61 percentage points. Validation loss also improved during controlled fine-tuning.



3. Evaluation, Comparison & Error Analysis

92.17%

FINAL TEST ACCURACY

92.35%

MACRO F1-SCORE

0.1959

FINAL TEST LOSS

Model Comparison

Model	Input	Trainable Params	Val. Acc.	Test Acc.	Macro F1
Baseline CNN	150×150×3	423,302	82.28%	82.30%	82.50%
Fine-Tuned MobileNetV2	224×224×3	1,202,566	92.78%	92.17%	92.35%

Transfer-learning gain: +10.50 percentage points in validation accuracy, +9.87 points in test accuracy, and +9.85 points in Macro F1 compared with the baseline CNN.

Per-Class Performance

Class	Precision	Recall	F1
Buildings	93.71%	91.99%	92.84%
Forest	98.94%	98.73%	98.84%
Glacier	86.86%	86.08%	86.47%
Mountain	87.18%	85.52%	86.35%
Sea	94.32%	97.65%	95.95%
Street	93.10%	94.21%	93.65%

Error analysis. The dominant confusion is **glacier** ↔ **mountain** (58 glacier→mountain and 62 mountain→glacier errors). A secondary urban confusion is **buildings** ↔ **street** (32 buildings→street and 26 street→buildings). These errors are concentrated in visually similar categories rather than being randomly distributed.

4. Explainability, Deployment & Final Conclusion

Grad-CAM Explainability

Grad-CAM was added to visualize the spatial regions that influence predictions. For external examples, the notebook presents the original image, activation heatmap, overlay, predicted class, and confidence. This makes the model more interpretable and helps verify that decisions are connected to meaningful scene regions.

Prediction on Completely New Images

A reusable prediction function accepts a new image path, displays the image, and returns both the predicted class and Softmax confidence. It was tested on six external images that were not used during training. Example predictions included street (74.92%), mountain (55.32%), street (95.85%), mountain (67.75%), mountain (82.47%), and forest (98.40%). Because these external prediction images are unlabeled, the confidence values are used for qualitative demonstration rather than accuracy measurement.

Deployment & Reproducibility

- The selected model is exported as `final_mobilenetv2_transfer_model.keras`.
- A professional **Streamlit** interface supports image upload, prediction, confidence display, and class probabilities.
- The project includes a **README.md** with setup, run, model-loading, preprocessing, and inference instructions.
- The final model already contains its required input rescaling, preventing accidental double preprocessing during deployment.

Production Recommendation

The **Fine-Tuned MobileNetV2** is the recommended production model. Although it has more trainable parameters than the baseline, it provides a large improvement in validation accuracy, test accuracy, and F1-score, while maintaining close validation-to-test performance. It also benefits from pretrained visual representations and integrates cleanly with the exported inference and Streamlit workflow.

Final Conclusion

This project delivers a complete end-to-end computer vision solution: data understanding and EDA, leakage-aware splitting, preprocessing and augmentation, baseline CNN development, transfer learning, controlled fine-tuning, quantitative evaluation, confusion analysis, Grad-CAM explainability, external-image prediction, model export, and interactive deployment. The final model achieves **92.17% test accuracy** and **92.35% Macro F1**, demonstrating that transfer learning substantially improves natural-scene recognition over the custom baseline.

Future Improvements

With additional time or data, the next improvements would include a larger and more diverse training set, targeted augmentation for glacier/mountain ambiguity, systematic hyperparameter optimization, probability calibration, comparison with EfficientNet or ResNet, and model compression or quantization for lightweight deployment.

FINAL STATUS

End-to-end classification pipeline completed with evaluation, explainability, export, and deployment components.